

Week 4 — Tongue Diagnosis

EAM310 · EAST ASIAN MEDICINE DIAGNOSIS I · VUIM SUMMER 2026

△ Quiz 3 — July 31, 2026

Class: Friday August 1, 2026 · 9AM–1PM

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REQUIRED READINGS — DUE BEFORE AUGUST 1

FCM CHAPTER 23 — MACIOCIA

- Tongue body color — normal, pale, red, purple
- Tongue body shape — swollen, cracked, deviated
- Tongue coating — color, thickness, moisture, root
- Sublingual veins
- Clinical significance of tongue diagnosis

FCM CHAPTER 24 — MACIOCIA + CAM CH.8

- Tongue body color in disease
- Coating color, viscosity & peeling patterns
- Topography — Three Burner mapping
- Zang-Fu tongue zones (Maciocia + Kirschbaum)
- Observation conditions & external factors

WEEK 4 LEARNING OBJECTIVES (CLO 1, 4, 5, 6)

- Identify the four main aspects of tongue diagnosis
- Interpret tongue body shape findings clinically
- Map tongue zones to Zang-Fu organs (Schrier + Maciocia)
- Apply medication effects on the tongue to clinical intake
- Distinguish normal from pathological tongue body color
- Decode tongue coating — color, thickness, moisture, viscosity
- Identify the three tongue layers and what each reflects
- Integrate tongue + pulse for complete pattern differentiation

ASSESSMENT CALENDAR

Date	Event	Material Covered
Jul 31	△ Quiz 3	Week 3 material — Pulse Diagnosis (28 qualities, normal pulse, positions)
Aug 1	Week 4 Class	Tongue Diagnosis — body color, shape, coating, topography, clinical cases
Aug 7	MIDTERM + Journal 1–25	Weeks 1–4 comprehensive
Sep 11	FINAL + Journal 26–50	Weeks 6–9 comprehensive

IN-CLASS PRACTICE — WHAT TO EXPECT FRIDAY

TONGUE OBSERVATION LAB

Week 4 includes live tongue observation practice with partners. Come ready to:

- Observe tongue body color and describe in TCM terms (pale, red, purple, etc.)
- Note coating — color (white/yellow/grey), thickness (thin/thick), moisture (wet/dry/sticky)
- Identify any shape features — swelling, cracks, tooth marks, deviation
- Document your observations in your Tongue/Pulse Journal (entry counts toward midterm)
- Bring your blank tongue map for annotation during observation

Why tongue?	Body color always shows the true condition — even when other signs mislead
①	Judge the exuberance/decline of antipathogenic Qi
②	Distinguish the thermal nature of a disease (Hot vs Cold)
③	Detect the location of a disease (zone mapping)
④	Predict the course of the disease

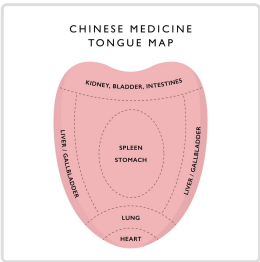
THREE TONGUE LAYERS — WHAT EACH REFLECTS		
Layer	Reflects	Depth
Coat 苔 / Wei	Stomach (Yang organs); pathogenic factors	Most superficial
Body 體 / Ying	Yin organs; Blood; Nutritive Qi — most reliable indicator	Middle
Sublingual / Yuan	Yuan Qi; Blood stasis (dark, distended, tortuous veins = stasis)	Deepest

TONGUE TOPOGRAPHY — ZONE MAPPING

Zone / Area	Organ(s)	Key Sign
Tip	Heart 心	Red tip = Heart-Fire or Heart Empty-Heat
Just behind tip	Lung 肺	Swollen front = Phlegm in Lungs
Sides	Liver / Gallbladder 肝胆	Red sides = LV-Fire (w/coat) or LV-Yin def (no coat)
Center oval	Spleen / Stomach 脾胃	Central crack = ST-Yin def
Root / Back	Kidney · BL · Intestines 腎	Rootless coat = KD-Yin def

FOUR MAIN ASPECTS OF TONGUE DIAGNOSIS	
Aspect	Indicates
1. Body Color	Blood, Nutritive Qi, Yin organs; Heat/Cold, Yin/Yang def — <i>always true condition</i>
2. Body Shape	Blood and Nutritive Qi; Excess or Deficiency
3. Coating	Yang organs, Stomach; Heat/Cold, Fullness/Deficiency
4. Moisture	State of Body Fluids; Yang Qi transformation
+ Spirit 神	Vitality — alive, lustrous vs dull and lifeless ("cow tongue at the kosher deli")

Color	Pale red — fits nicely inside mouth
Shape	Neither swollen, thin, cracked, loose nor moving when extended
Coating	Thin white coat — slightly moist, not too dry, not too wet
Spirit	Alive, lustrous, vibrant



Tongue Zone Map (Maciocia / Sacred Lotus)

Contralateral rule: Left tongue side = right breast; Right tongue side = left breast (Kirschbaum)

OBSERVATION CONDITIONS & EXTERNAL FACTORS

Factor	Rule
Lighting	Natural light only — fluorescent distorts color. Classroom fluorescent = unreliable.
Sequence	Tip → middle → root. 5–10 seconds per observation. Max 15–20 seconds total — prolonged extension changes moisture artificially.
Diet & Meds	Coffee, tea, colored food, smoking throw off coat. Nothing coloring 30 min before.
Season / Time	Tongue shifts through seasons. Factor in on borderline findings.
Age / Constitution	Children: generally redder (normal). Elderly: drier, paler, more contracted (normal).

Color	Indicates	Clinical Details — From Slides
PALE		
Pale	Yang def; Blood def; Cold	Pale + wet = Yang def · Pale + dry = Blood def · Pale sides = LV-Blood def · Pale + slightly red tip = HT-Blood def with some Heat
RED		
Red (w/coat)	Full-Heat	Red + yellow coat = Full-Heat · Red tip + coat = HT-Fire · Red sides + coat = LV-Fire or GB Full-Heat · Red center + coat = ST Heat
Red (no coat)	Empty-Heat	Red + peeled = Yin def with Empty-Heat · Red tip no coat = HT Empty-Heat · Red sides no coat = LV-Yin def with Empty-Heat
Deep Red / Scarlet	Severe Heat	Deeper red = more severe; may indicate Ying or Blood level heat in febrile disease
PURPLE		
Reddish-Purple	Blood stasis from Heat	Blood stasis + Heat · Reddish-purple sides = LV-Blood stasis with Heat · Purple center = Blood stasis in Stomach
Bluish-Purple	Blood stasis from Cold	Blood stasis + Cold · Bluish-purple + wet = Cold stagnating Blood · Purple with dark sublingual veins = severe Blood stasis
★ Red Points Test (Schrier quiz tip): Red points on the tip = HT-Fire · Red points on the sides = LV-Fire · Red spots on the root = KD/Lower Jiao Heat · Red spots scattered = Nutritive Qi Heat		
BODY SHAPE — 舌形		
Shape Finding	Indicates	Key Notes
Swollen (overall)	Dampness; Phlegm; Yang def	Pale + swollen + wet = Yang def with Cold-Damp · Red + swollen = Damp-Heat or Phlegm-Heat
Swollen sides	LV-Fire or LV-Qi stagnation	Red swollen sides = LV-Fire · Pale swollen sides = LV-Blood def
Swollen tip (red)	HT-Fire	Can also indicate Lung area swelling = LU-Qi def if pale and swollen in front third
Tooth-Marked	Spleen-Qi def; Dampness	Almost always Spleen-Qi deficiency — the swollen tongue presses against the teeth
Cracked / Fissured	Yin def; Heat injuring fluids	Central crack = ST-Yin def or ST-Heat · Transverse cracks = SP-Yin def · Many fine cracks = Blood or Essence def · Normal in some people (check history)
Thin / Narrow	Blood def; Yin def	Thin + pale = Blood def · Thin + red, no coat = Yin def with Empty-Heat
Deviated	Interior Wind	Always check for mechanical/trauma cause first — Schrier: "I got punched in high school." Deviating to one side on protrusion = LV-Wind
Moving / Trembling	Liver Wind	Involuntary movement when extended · SP-Qi def (pale + trembling) · LV-Wind (red + trembling)
Stiff / Hard	Heat disturbing Mind; Wind	Inability to move freely · Heat in Nutritive Qi or Pericardium · Interior Wind
Flaccid / Limp	Severe Qi/Blood def; Yin def	Tongue cannot move · Pale + flaccid = severe Qi/Blood def · Red + flaccid, no coat = severe Yin def
Long	Heat in HT; Phlegm-Fire	Tongue extends beyond the lips · Red + long = HT-Fire or Phlegm-Fire
Short / Contracted	Critical condition	Pale + short = Cold contracting sinews · Red/purple + short = Heat injuring sinews · Always serious

		COATING THICKNESS	
Color	Indicates	Thin coat	Normal, early stage, or mild pathogen — can see tongue through it
White Thin-white	Normal; exterior Cold; early-stage exterior · Thin = mild or early; Thick = more pathology	Thick coat	Stronger pathogenic factor; the thicker = the stronger the pathogen
White Thick-white	Interior Cold-Damp; food stagnation · Thick + wet = Damp-Cold in interior	No coat	ST-Yin def (coat fails to form) · Total absence + red = ST & KD Yin def with Empty-Heat
Yellow Thin-yellow	Exterior Wind-Heat (early stage); Wind-Cold transforming to Heat		
Yellow Thick-yellow	Interior Full-Heat · Damp-Heat accumulating with Qi stagnation		
Yellow + Greasy	Damp-Heat or Phlegm-Heat · Food stagnation with Heat	Slightly moist	Normal — Body Fluids intact, Yang Qi transforming properly
Grey (Glossy)	Cold-Damp or Damp-Phlegm (interior Cold)	Wet / Glossy	Yang def — fluids not transformed · Cold-Damp · Water accumulation
Grey (Dry)	Yin deficiency with Heat signs	Dry	Heat or Yin def damaging fluids · Also Dryness pathogen
Black (Wet)	Extreme Cold · Severe Yang def with interior Cold	Sticky / Slippery	Dampness and/or Phlegm — coating like an oily film
Black (Dry)	Extreme Heat · Yin def with Empty-Heat (severe)		
		COATING VISCOSITY	
		Greasy coat	Sticky fine particles thicker in middle, thinner at margins — Dampness/Phlegm
		Rough coat	Granules rough like sandpaper — Heat injuring fluids
		Moldy coat	Thick, cheesy, cottage-cheese-like — Turbid Dampness rising
COATING PEELING PATTERNS			
Pattern	Description	Indicates	
Partially peeled	Scattered areas of peeled tissue with borders rising above surface	ST-Yin and Qi def	
Geographic	Large irregular areas missing — resembles a map	ST/SP Yin def	
Mirror / Smooth	Lacks all coating; looks smooth and shiny	Stomach Yin exhausted — serious	
Dappled	Portions missing; surface not smooth; new coat growing	ST-Yin and Qi def recovering	
MEDICATIONS & EFFECT ON TONGUE (ALL 7 — FROM SLIDES)			
Medication	Tongue Effect	Clinical Pearl	
Antibiotics	Peeled patches	"Are you on antibiotics?" = instant credibility. No Yin def signs → medication artifact	
Corticosteroids	Red, swollen; tip much redder	Mimics Damp-Heat or Full-Heat	
Bronchodilators	Red tip only — after many years	Long-term inhalers → Heart/Lung heat at tip	
Diuretics	Peeling	Dries Body Fluids → ST/Yin deficiency appearance	
NSAIDs	Red points; thin body	Long-term NSAIDs generate internal Heat	
Chemotherapy	Very thick brown/black dry coating	Distinguish from Toxic-Heat	
Pepto-Bismol	Black tongue	Bismuth + oral sulfur. Harmless. Ask immediately. Mystery solved.	

Tongue Finding	Likely Pattern
Slightly pale body	Qi deficiency
Pale + dry	Blood deficiency
Pale on sides	Liver-Blood deficiency
Pale + slightly wet	Yang deficiency
Tooth-marked / swollen center	Spleen-Qi deficiency
Rootless coating	Stomach-Qi/Yin deficiency
No coating (all over)	Yin deficiency
No coating in center	Stomach-Yin deficiency
No coating in front third	Lung-Yin deficiency
No coating all over + red body	KD + ST Yin def with Empty-Heat
Red body + yellow coat	Full-Heat
Red body + no coat	Empty-Heat (Yin def)
Red tip + coat	Heart-Fire
Red tip + no coat	Heart Empty-Heat
Red sides + coat	Liver-Fire
Red sides + no coat	Liver-Yin def with Empty-Heat
Purple + wet	Blood stasis from Cold
Purple + dry	Blood stasis from Heat
Dark sublingual veins	Blood stasis (any cause)
Cracked center + red	ST-Heat / Yin def
Deviated tongue	Interior Wind (check trauma first)
Yellow coat = Heat · Deviated = Int Wind	Miley Cyrus clinical pearl (Schrier)
Thick white coat + wet	Cold-Damp interior
Yellow greasy coat	Damp-Heat or Phlegm-Heat
Black dry coat	Extreme Heat / Yin def severe
Black wet coat	Extreme Cold / severe Yang def

WIND-COLD VS WIND-HEAT — TONGUE PROGRESSION

Stage	Tongue Pattern
Wind-Cold	Thin white coat (exterior not yet interior)
Wind-Heat (early)	Thin white coat → beginning of yellow
Wind-Heat (interior)	Yellow coat; body becomes redder
True Cold / False Heat	Pale body + black wet coat
Cold transformed to Heat	White coat → yellow coat progression

Normal	Fine, blue-ish, not distended
Dark & distended	Blood stasis — most reliable sign
Tortuous / varicose	Severe Blood stasis; long-standing
Bright red veins	Heat in Blood or Ying level

COVID-19 / SARS TONGUE PATTERNS (SCHRIER BONUS SLIDES)

From Ioannis Solos article (Wk 4 slides 21–22):

- **Red tip + white root:** External Heat with internal Cold — early COVID presentation
- **White fur transmuting to yellow:** Exterior pathogen penetrating to interior — watch the coat transform
- **Light red body + black/gray patches:** Heat with Blood stasis toxin
- **Old yellow fur + petals:** Severe Damp-Heat accumulation

△ Bonus/not tested per slides. Clinical reference only.

★ **MIDTERM BRIDGE NOTE:** Week 4 Tongue Diagnosis closes the Weeks 1–4 midterm block. The midterm (Aug 7) covers Weeks 1–4 comprehensively — all tongue zones, colors, coatings, shapes, and their pattern correspondences are fair game.

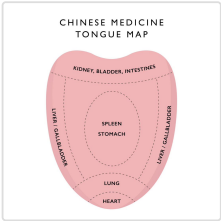
EAM Diagnosis Notes

EAM310 · EAM Diagnosis I · Prof. Schrier · Friday 9AM–1PM · Room 303

PATTERN NAME	DISEASE CATEGORY	WEEK / QUIZ
ETIOLOGY	ORGAN SYSTEM	DATE

CLINICAL ESSENTIALS · TONGUE · PULSE · TREATMENT PRINCIPLE

TONGUE 舌診



COLOR / COAT / SHAPE

PULSE 脈診 · CUN · GUAN · CHI

Wrist	CUN 寸	GUAN 關	CHI 尺
L HT/PC		LV/GB	KD-Yin
R LU/LI		SP/ST	KD-Yang

QUALITY / OVERALL

TREATMENT PRINCIPLE 治則

KEY POINTS / HERBS

